FLIGHT ENDURANCE

OVERVIEW

Participants apply leadership and/or 21st century skills during the design iteration process in which participants build, fly, and adjust (trim) a rubber-band powered model aircraft to make long endurance flights inside a contained airspace. Models must be of fixed-wing design and comply with all event specifications.

ELIGIBILITY

Two (2) individuals per chapter may participate.

TIME LIMITS

Thirty (30) minutes is allowed to trim flights.

ATTIRE

TSA competition attire is required for this event.

PROCEDURE

PRE-CONFERENCE

A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules.

B. Participants review the Use of Generative Artificial Intelligence (GenAI) found in the General Rules.

ON-SITE TESTING OF PRE-BUILT AND TRIMMED AIRCRAFT

A. Participants check in the following at the time and place stated in the conference program:

   1. Completed aircraft
   2. Portfolio
   3. Safety glasses

B. Aircraft will be stored in a secure holding area until the scheduled time for trim and official flights.

ON-SITE TESTING

A. Prior to trim flying, participants are strongly encouraged to check out the designated competition room for this event.

B. Participants attend a pilot's meeting to review the sequence for making official flights.

C. Participants arrive at the competition site for trim flying during the time designated for their heat.

D. Time allotted for the trim portion of the event may be extended according to the number of participants and site scheduling.

E. Participants have two (2) opportunities to fly their models for official times.

OFFICIAL FLIGHT TEST

A. In an orderly fashion, participants proceed to a group timer for permission to fly.

B. Participants place their models on the floor and wait for the release signal from the timer. Timing begins when the model rises off the ground.

C. Flight time ends at the moment the model first touches the floor/ground, or the moment it comes to rest against any surface or obstruction.

D. Timekeeper records the two (2) official flight times and landing bonus(s) for each participant.

E. Immediately following the second flight, the participant hands his/her motor to the judge for weighing. Place in a zip lock bag and record the entry number on the outside.

SCORING

A. Entries are evaluated by the judges with neither students nor advisors present based on the following criteria:

   1. Judges enter the flight times and landing bonus(s) for each entry to determine the top sixteen (16) contestants, which will not be posted.
   2. Judges complete the inspection of the model and flight box, enter the placement points, and judge the documentation portfolio to determine the ten (10) finalists.

B. A model or flight box that fails inspection will receive a zero (0) for the flight rankings.

C. The top ten (10) finalists are announced at the awards ceremony.

Flight Log:

Participant ID#:			Dates:	
Flight #	# of winds	Time aloft	Flight pattern	Trim adjustment
#1				
#2				
#3				
#4				
#5				
#6				
#7				
#8				
#9				
#10				

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

A. Flight Endurance is an individual event.
   1. No one may assist the participant in any way during either trim or official flights.
   2. Violation of this regulation will result in disqualification.

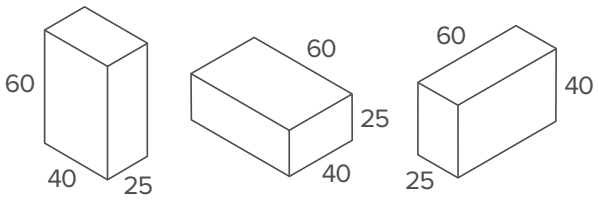
B. Documentation Portfolio:
   1. Documentation materials (comprising "a portfolio") are required and must be secured in a clear front report cover with the following single-sided, 8½" x 11" pages, in this order:
      a. Title page with the event title, the member identification number, the conference city and state, and the year; one (1) page
      b. A flight log; pages as needed (see official sample above), with the previous ten (10) flights.
      c. Technical attributes of the design and a description and identification of parts; pages as needed
      d. An analysis of the modifications and an explanation of why each was made must be included; pages as needed

e. A technical review of the flight log that explains the trim adjustments and modifications required to improve endurance. Experts from the Academy of Model Aeronautics (AMA) and the National Free Flight Society (NFFS) may scrutinize this information for validity; one (1) page

f. Assembly drawing of plane (full size or scale drawing); pages as needed

g. Scaled engineered drawings of all structural parts of the plane; pages as needed

C. Model and its parts must be contained in a flight box that does not exceed 25cm x 40cm x 60cm. Flight box hardware, such as hinges, handles, and wheels, are not measured.



D. **A flight box that violates any part of Regulation C will disqualify the model.**

E. Models:

1. Models are to be made of any materials that are typically found in model construction. This includes, but is not limited to: wood, foam, foam board, and plastics.
   a. Hardeners are permitted but are not required.
   b. **Use of any materials that are deemed unsafe will not be tested and will be disqualified.** Aircraft containing Boron filaments/fibers shall be disqualified and will not be tested.
2. Models must use a fixed-pitch propeller with a maximum of 170mm diameter.
   a. Propellers may be trimmed, shaped, balanced, or re-pitched, but must remain fixed in pitch.
   b. Variable-pitch propellers and/or mechanisms are NOT permitted.
3. Rotary-wing aircraft and aerostat (lighter than air) aircraft are NOT permitted.
4. Wingspan: maximum 40cm (horizontal projection). Maximum projected span of all lifting surfaces, from farthest left tip to farthest right tip
5. Wing chord of each lifting surface: maximum 9cm (perpendicular projected chord).
6. Rubber motor: maximum weight of motor is 1.50 grams, including the O-rings.
   a. No length measurement is made.
   b. Spare motors are allowed during the official flights.
   c. Two (2) rubber O-rings may be used on the rubber motor loop for easier handling of wound motors.
7. Model weight: minimum of 8.0 grams.
   a. Models are weighed without motors attached.
   b. Ballast is permitted for trimming and ballast.
   c. Model is weighed with ballast.
8. Steel wire may be used only for the propeller shaft, motor hook, landing gear, and the connection between fuselage and tail. Small plastic tubes, such as coffee stirrers, may be used.
9. The two (2) wheels must be a minimum of 15mm in diameter, and they must roll freely by the weight of the plane on a smooth surface.
10. When at rest, the landing gear must support the model without the fuselage and/or propeller touching the floor or launching pad.

F. Acceptable flight support equipment includes the following:

1. Mechanical rubber motor winders or battery-powered motor winders may be used. No AC-powered winders are allowed.
2. A winding stooge may be used to anchor the model while the motor is being wound. A person may not serve as a winding stooge.
3. A poster board launching platform is provided.

G. Only minor repairs are allowed during trim and time trials.

EVALUATION

PRELIMINARY ROUND

Tier 1

A. Flight times and landing bonus(s)

Tier 2

B. Inspection of model and flight box

C. Flight ranking points

D. Documentation portfolio

Refer to the official rating form for more information.

GENERATIVE ARTIFICIAL INTELLIGENCE

Use of Generative Artificial Intelligence (GenAI) is NOT permitted for this event.

NOTES

Two organizations—the Academy of Model Aeronautics (AMA) and the National Free Flight Society (NFFS)—welcome your inquiries and offer suggestions, help, and technical information concerning model aircraft and flight technology.

Contact the AMA: www.modelaircraft.org.

Contact NFFS: www.freeflight.org.

STEM INTEGRATION

This event aligns with the STEM (Science, Technology, Engineering, and Mathematics) educational standards.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Aeronautical engineer
- Aircraft systems engineer
- Physics teacher

Participant/Team ID# _____

FLIGHT ENDURANCE

2027 & 2028 OFFICIAL RATING FORM

HIGH SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ TIER 1 – Model is present for flight times
☐ TIER 2 – Inspection, flight rankings, and documentation portfolio
☐ ENTRY NOT EVALUATED

TIER 1 – FLIGHT TIMES		
Flight times recorded to the nearest tenth (.1) of a second.		
Duration of Flight #1	Seconds	
Duration of Flight #2	Seconds	
Landing Bonus – add ten (10) seconds for each successful landing	Seconds	
TIER 1 – TOTAL FLIGHT TIMES (combine flight #1, flight #2, and landing bonus)	**SECONDS**	

TIER 2 – FLIGHT RANKINGS (60 points)								
1st	2nd	3rd	4th	5th & 6th	7th & 8th	9th-12th	13th – 16th	
60 Points	55 Points	50 Points	45 Points	40 Points	35 Points	30 Points	20 Points	
TIER 2 – FLIGHT RANKINGS (60 points)								

TIER 2 – DOCUMENTATION PORTFOLIO (70 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Portfolio Components (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio has most components and is generally organized per the order of the event guide listing; it has sufficient content.	All components are included in the portfolio; content and organization per the order of the event guide listing are excellent.	
Flight Log (X1)	Flight log is incomplete.	Flight log is generally complete.	Flight log is complete; a thorough understanding of the flight log's purpose is evident.	

TIER 2 – DOCUMENTATION PORTFOLIO (70 points) – continued				
Technical Attributes (X1)	Attributes of the design reflect no knowledge of flight design.	Attributes of the design are included and adequately reflect basic knowledge of flight design.	Clear and precise attributes of the design are given; an in-depth knowledge of flight design is exhibited.	
Analysis of the Modifications and Explanation (X1)	Only one (1) modification is noted, and/or an explanation of why the modification was made is missing; leadership and/or 21st century skills are not evident.	Modifications are given with adequate explanations for how they improved flight endurance; leadership and/or 21st century skills are somewhat evident.	Modifications and an explanation of why they were made are provided; a clear and precise explanation for how they improved the flight endurance is provided; leadership and/or 21st century skills are clearly evident.	
Technical Review of Flight Log (X1)	Only one (1) review of trim adjustments or modifications are included to improve endurance.	Most reviews of trim adjustments or modifications are included to improve endurance.	Review of the flight log includes precise multiple explanations of trim adjustments and modifications to improve endurance.	
Assembly Drawing (X1)	Assembly drawing is unclear; majority of the design principles are not addressed or are missing; pictures are missing.	Assembly drawing is partially clear; most of the design principles are addressed and/or present; some pictures are missing.	Assembly drawing is clear, accurate, and executed well; all design principles are addressed; no pictures are missing.	
Scaled Engineering Drawings (X1)	Majority of the parts are not described, sourced, or identified accurately; scaled engineered drawings are incomplete or missing.	Most parts are described and sourced accurately; scaled engineered drawings include most details.	All parts are described and sourced completely and accurately; engineering drawings are complete.	
			TIER 2 – DOCUMENTATION PORTFOLIO SUBTOTAL (70 points)	

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. **TOTAL** (130 points)

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

FLIGHT ENDURANCE
EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

A. Event coordinator
B. Assistants, two (2) or more
C. Judges, two (2) or more
D. Timekeepers, two (2)

MATERIALS

A. Coordinator's packet, containing:
    1. Event guidelines, one (1) copy for the coordinator and for each judge
    2. TSA Event Coordinator Report
B. Marking pens (sharpie and felt tip. fine point)
C. Two (2) metric tape measures
D. Two (2) rolls of caution tape
E. 125 zip lock bags (for motor storage for weighing)
F. Three (3) launch pads (poster board, 30" x 40")
G. Signs for door(s) reading Do Not Open, Flight in Progress, Knock for Entry
H. Stopwatches, three (3)
I. Electronic gram scale (to .01 gram)

RESPONSIBILITIES

AT THE CONFERENCE

A. Attend the mandatory coordinator's meeting at the designated time and location.
B. Report to the CRC room and check the contents of the coordinator's packet.
C. Review the event guidelines and check to see that enough judges and assistants have been scheduled.
D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, tables, chairs, etc. Notify the event manager of any potential problems.
E. At least one (1) hour before the event is to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
F. Ensure the judges have access to the online judging system.

EVENT CHECK-IN

A. Check in participants and evaluate models for special compliance during the scheduled trim session.
B. If check-in is on the first day of the conference, requirements for attire do NOT apply during check-in.
C. Anyone reporting who is not on the entry list may check in only after official notification is received from the CRC.
D. Late entries are considered on a case-by-case basis and only when the delay is caused by events beyond participant control.
E. Secure aircraft in the holding area so that they remain safe until the scheduled time for the official flights.

PRELIMINARY ROUND

A. Following the pilot's meeting, each entry is allowed to be flown and times recorded.
B. Up to three (3) groups may fly simultaneously in the assigned area for the event, with consideration for the safety of the models and participants.
C. Distribute a list of entrants assigned to each designated judge/timer.
D. Each flight is recorded to the nearest one-tenth (.1) of a second and landing bonus(s) are recorded.
E. Entries are evaluated by the judges with neither students nor advisors present based on the following criteria:
    1. Judges enter the flight times and landing bonus(s) for each entry to determine the top sixteen (16) contestants, which will not be posted.
    2. Judges complete the inspection of the model and flight box, enter the placement points, and judge the documentation portfolio to determine the ten (10) finalists.

F. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and CRC manager to determine either:

   1. to deduct twenty percent (20%) of the total possible points in this round or
   2. to disqualify the entry

   The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

G. If necessary, manage security and the removal of materials from the event area.